

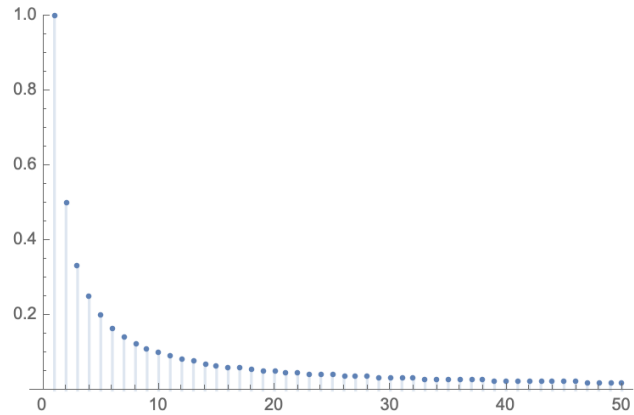
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Summation of series

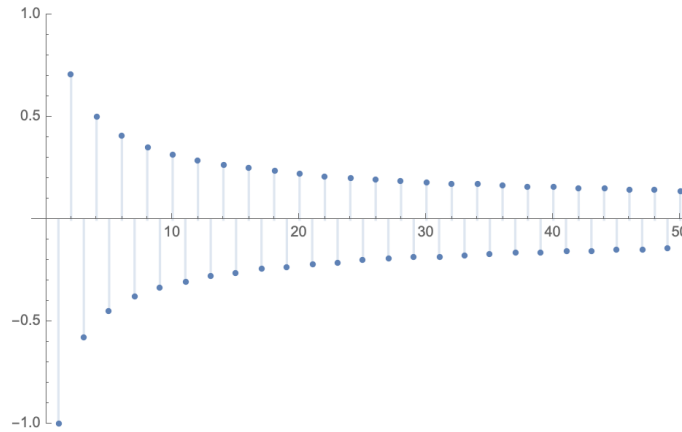
Infinite sequences

An infinite sequence $\{a_n\}$ of real numbers is a function whose domain is $\mathbb{N}$.

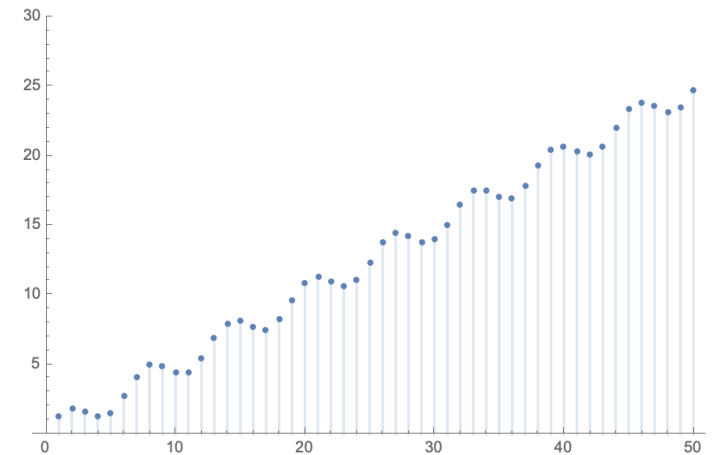
`DiscretePlot[1/n, {n, 1, 50}, PlotRange -> {0, 1}]`



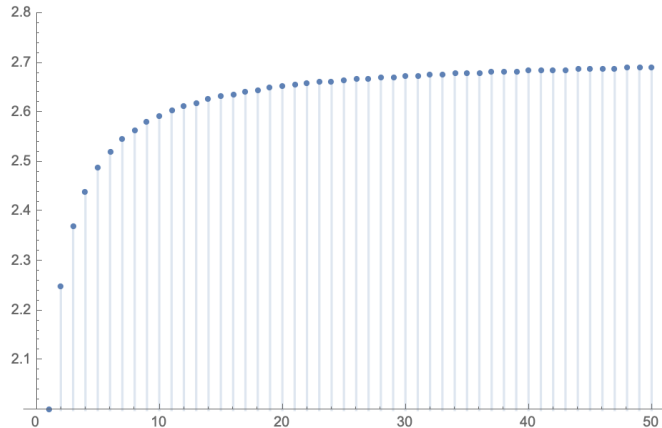
`DiscretePlot[(-1)^n / Sqrt[n], {n, 1, 50}, PlotRange -> {-1, 1}]`



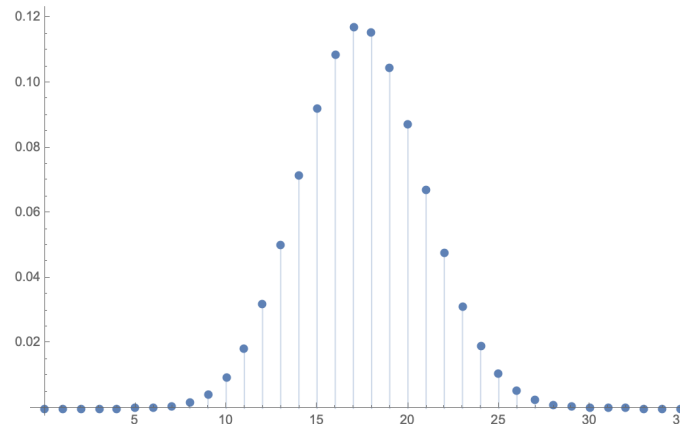
`DiscretePlot[n/2 + Sin[n], {n, 1, 50}, PlotRange -> {0, 30}]`



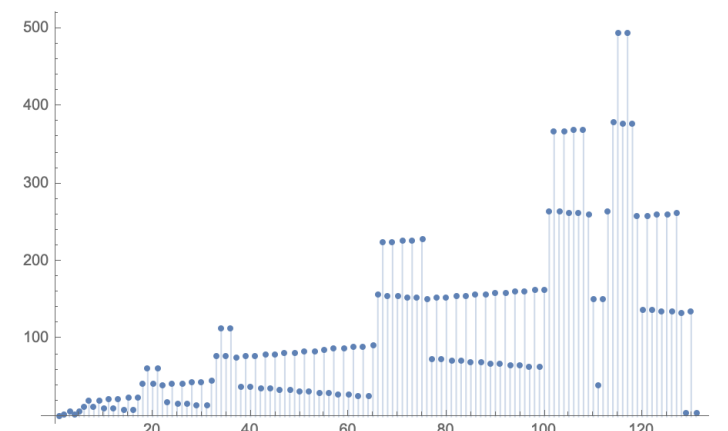
`DiscretePlot[((n+1)/n)^n, {n, 1, 50}, PlotRange -> {2, 2.8}]`



`ListPlot[Table[{k, PDF[BinomialDistribution[50, p], k]}, {p, {0.35}}, {k, 0, 35}], Filling -> Axis]`



`ListPlot[a, Filling -> Axis]`



Series

A **series** is a sequence of **partial sums**.

Arithmetic sequences

The sum of n terms of an arithmetic sequence, starting term a with common difference d is

$$\frac{n}{2} [2a + (n - 1)d]$$

$1, 2, 3, 4, 5, 6, \dots \dots n, \dots \rightarrow +\infty$ as $n \rightarrow \infty$

$1, 1 + 2, 1 + 2 + 3, 1 + 2 + 3 + 4, \dots \dots \dots, 1 + 2 + 3 + \dots \dots + n, \dots \rightarrow +\infty$ as $n \rightarrow \infty$

Sum of the first n naturals

$$(r + 1)^2 - r^2 \equiv (r + 1 - r)(r + 1 + r)$$

$$\Rightarrow \sum_{r=1}^n [(r + 1)^2 - r^2] = \sum_{r=1}^n (2r + 1)$$

$$\Rightarrow (n + 1)^2 - 1 = 2 \left(\sum_{r=1}^n r \right) + n$$

$$\Rightarrow \sum_{r=1}^n r = \frac{1}{2} n (n + 1)$$

Arithmetic sequences

The sum of n terms of an arithmetic sequence, starting term a with common difference d is

$$\frac{n}{2} [2a + (n - 1)d]$$

$$\sum_{r=1}^n r = \frac{1}{2} n(n + 1)$$

1, 2, 3, 4, 5, 6, n , ... $\rightarrow +\infty$ as $n \rightarrow \infty$

1, 1 + 2, 1 + 2 + 3, 1 + 2 + 3 + 4,, 1 + 2 + 3 + + n , ... $\rightarrow +\infty$ as $n \rightarrow \infty$

1, 5, 9, 13, 17, 21,, $1 + (n - 1)4$, ... $\rightarrow +\infty$ as $n \rightarrow \infty$ $\rightarrow +\infty$ as $n \rightarrow \infty$

1, 1 + 5, 1 + 5 + 9, 1 + 5 + 9 + 13,, 1 + 5 + 9 + + $1 + (n - 1)4$, ...

$$\sum_{r=1}^n (1 + 4(r - 1)) = \sum_{r=1}^n (4r - 3) = ???$$

Geometric sequences

The sum of n terms of a geometric sequence, denoted S_n , having starting term a with common ratio r is $\frac{a(r^n - 1)}{r - 1}$ or $\frac{a(1 - r^n)}{1 - r}$ where $r \neq 1$.

If $|r| < 1$ then the infinite sum is

$$\frac{a}{1 - r}.$$

$1, 2, 4, 8, 16, 32, \dots \dots 2^{n-1}, \dots \rightarrow +\infty$ as $n \rightarrow \infty$

$$\sum_{r=1}^n 2^{r-1} = 2^n - 1$$

$1, 1 + 2, 1 + 2 + 4, 1 + 2 + 4 + 8, \dots \dots \dots, 1 + 2 + 4 + \dots \dots + 2^{n-1}, \dots$

$\rightarrow +\infty$ as $n \rightarrow \infty$

$\frac{1}{1}, \frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \frac{1}{81}, \dots \dots, \frac{1}{3^{n-1}}, \dots \rightarrow 0$ as $n \rightarrow \infty$

$$\sum_{r=1}^n \frac{1}{3^{r-1}} = \frac{3}{2} \left(1 - \frac{1}{3^n} \right)$$

$\frac{1}{1}, \frac{1}{1} + \frac{1}{3}, \frac{1}{1} + \frac{1}{3} + \frac{1}{9}, \dots \dots \dots, \frac{1}{1} + \frac{1}{3} + \frac{1}{9} + \dots \dots + \frac{1}{3^{n-1}}, \dots$

$\rightarrow \frac{3}{2}$ as $n \rightarrow \infty$

$$\sum_{r=1}^{\infty} \frac{1}{3^{r-1}} = \frac{3}{2}$$

Sum of the first n squares

$$(r + 1)^3 - r^3 \equiv (r + 1 - r)[(r + 1)^2 + (r + 1)r + r^2]$$

$$\Rightarrow \sum_{r=1}^n [(r + 1)^3 - r^3] = \sum_{r=1}^n (3r^2 + 3r + 1)$$

$$\Rightarrow (n + 1)^3 - 1 = 3 \left(\sum_{r=1}^n r^2 \right) + 3 \left(\sum_{r=1}^n r \right) + n$$

$$\Rightarrow \sum_{r=1}^n r^2 = \frac{1}{6}n(2n + 1)(n + 1)$$

Some standard formulae

$$\sum_{r=1}^n r = \frac{1}{2}n(n+1)$$

$$\sum_{r=1}^n r^2 = \frac{1}{6}n(n+1)(2n+1)$$

$$\sum_{r=1}^n r^3 = \frac{1}{4}n^2(n+1)^2 = \left(\sum_{r=1}^n r \right)^2$$

WORKED EXAMPLE 3.1

Find $\sum_{r=1}^n (4r + 1)$. Hence determine the value of $\sum_{17}^{48} (4r + 1)$.

Answer

$$\sum_1^n (4r + 1) = 4 \sum_1^n r + \sum_1^n 1$$

Split the sum into smaller parts.

$$= 4 \times \frac{n(n+1)}{2} + n$$

Use the sum of natural numbers.

$$= 2n^2 + 3n$$

Combine terms.

$$\therefore \sum_{17}^{48} (4r + 1) = (2 \times 48^2 + 3 \times 48) - (2 \times 16^2 + 3 \times 16)$$

Write out as the difference of two separate sums, one with limits of 1 and 16, the other with limits of 1 and 48.

$$= 4192$$

Determine the final answer.

We can use $\sum r$ instead of $\sum_{r=1}^n r$ provided we remember the correct limits.

WORKED EXAMPLE 3.2

Find an expression for $1 + 3 + 5 + 7 + \dots$, for the first n terms, using two different approaches.

Answer

Method 1:

$$1 + 3 + 5 + 7 + \dots = \sum_{r=1}^n (2r - 1)$$

Write the sum of n odd terms.

$$= 2 \sum_{r=1}^n r - n$$

Use the standard result.

$$= n(n + 1) - n$$

Simplify.

$$\Rightarrow 1 + 3 + 5 + 7 + \dots = n^2$$

Method 2:

$$1 + 3 + 5 + 7 + \dots = \sum_{r=1}^{2n} r - \sum_{r=1}^n (2r)$$

Write a sum of $2n$ terms and subtract all even terms, leaving n terms.

$$= \frac{1}{2}(2n)(2n + 1) - 2 \times \frac{1}{2}n(n + 1)$$

The second sum is twice the sum of the first n natural numbers.

$$= n^2$$

This gives the same result.

WORKED EXAMPLE 3.3

Find an expression in terms of n for $\sum_{r=1}^n (3r^2 - 4r + 2)$.

Answer

$$\sum_{r=1}^n (3r^2 - 4r + 2) = 3 \sum_{r=1}^n r^2 - 4 \sum_{r=1}^n r + \sum_{r=1}^n 2$$

Split into individual sums.

$$= 3 \times \frac{1}{6}n(n+1)(2n+1) - 4 \times \frac{1}{2}n(n+1) + 2n$$

Use the result for each sum.

$$= \frac{n}{2}(2n^2 + 3n + 1) - 2n^2 - 2n + 2n$$

$$= n^3 + \frac{3}{2}n^2 + \frac{1}{2}n - 2n^2$$

$$= n^3 - \frac{1}{2}n^2 + \frac{1}{2}n$$

Simplify.

WORKED EXAMPLE 3.4

Find an expression in terms of n for $\sum_{r=1}^n r^2(r-1)$.

Answer

$$\sum_{r=1}^n r^2(r-1) = \sum_{r=1}^n r^3 - \sum_{r=1}^n r^2$$

Write as two separate sums.

$$= \frac{1}{4}n^2(n+1)^2 - \frac{1}{6}n(n+1)(2n+1)$$

Replace with the appropriate formulae.

$$= n(n+1) \left(\frac{n^2}{4} + \frac{n}{4} - \frac{n}{3} - \frac{1}{6} \right)$$

Note we do not need to expand all the brackets to simplify.

$$= \frac{1}{12}n(n+1)(3n^2 - n - 2)$$

$$= \frac{1}{12}n(n+1)(n-1)(3n+2)$$

Simplify.

EXERCISE 3A

P 1 Show that $\sum_{r=1}^n (3r - 5) = \frac{n}{2}(3n - 7)$.

P 2 Prove that $\sum_{r=1}^n 2r(r - 3) = \frac{2}{3}n(n + 1)(n - 4)$.

PS 3 Determine the value of $1 - 2 + 3 - 4 + \dots + (2n - 1) - (2n)$.

 **P** 4 Without using a calculator, show that $\sum_{r=1}^n (r + 2)(r + 3) = \frac{n(n^2 - 19)}{3}$.

 **PS** 5 Without using a calculator, find a simplified expression for $1^2 + 2^2 - 3^2 + 4^2 + 5^2 - 6^2 + \dots + (3n - 2)^2 + (3n - 1)^2 - (3n)^2$.

PS 6 Find, in terms of n , an expression for $\sum_{r=n+1}^{2n} r(r + 1)$.

M 7 Find the sum of the terms $1^2 + 2^2 + 3^2 + 5^2 + 6^2 + 7^2 + \dots + 97^2 + 98^2 + 99^2$.

P PS 8 Show that $\sum_{r=1}^n (2r + 1)(3r + 2) = \frac{n(4n^2 + 13n + 13)}{2}$. Hence, find the result of $\sum_{r=11}^{25} (2r + 1)(3r + 2)$.

M PS 9 Find the sum of the cubes of the first 100 even terms.

Some (surprising) reciprocal sums

$$\frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots, \frac{1}{n}, \dots \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$\frac{1}{1}, \quad \frac{1}{1} + \frac{1}{2}, \quad \frac{1}{1} + \frac{1}{2} + \frac{1}{3}, \dots, \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}, \dots \rightarrow ?? \text{ as } n \rightarrow \infty.$$

$\rightarrow +\infty \text{ as } n \rightarrow \infty$

$$\frac{1}{1^2}, \frac{1}{2^2}, \frac{1}{3^2}, \frac{1}{4^2}, \frac{1}{5^2}, \dots, \frac{1}{n^2}, \dots \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$\frac{1}{1^2}, \quad \frac{1}{1^2} + \frac{1}{2^2}, \quad \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2}, \dots, \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2}, \dots \rightarrow ?? \text{ as } n \rightarrow \infty.$$

$$\sum_{r=1}^{\infty} \frac{1}{r^2} = \zeta(2) = \frac{\pi^2}{6} \approx 1.64493$$

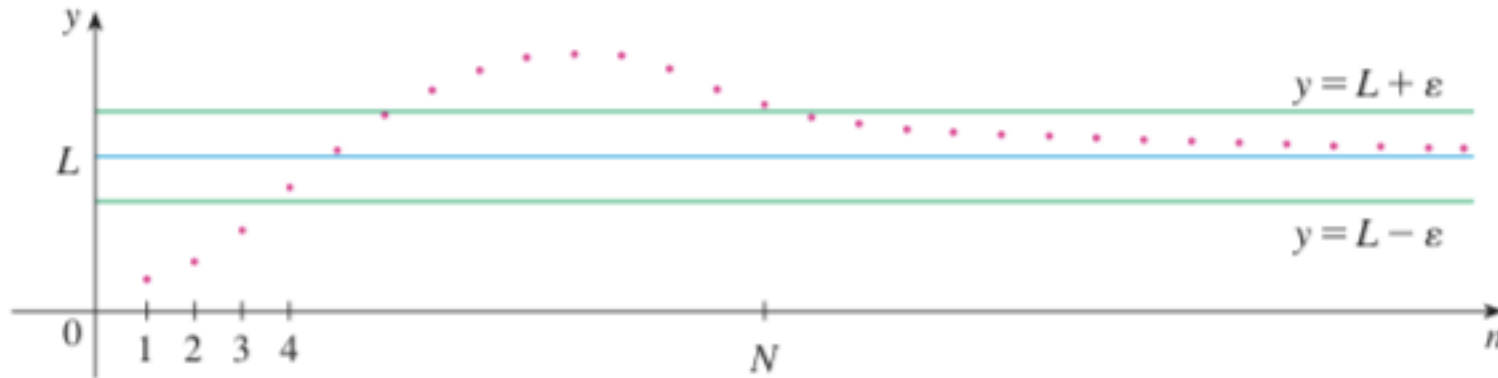
$$\frac{1}{1^3}, \frac{1}{2^3}, \frac{1}{3^3}, \frac{1}{4^3}, \frac{1}{5^3}, \dots, \frac{1}{n^3}, \dots \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$\frac{1}{1^3}, \quad \frac{1}{1^3} + \frac{1}{2^3}, \quad \frac{1}{1^3} + \frac{1}{2^3} + \frac{1}{3^3}, \dots, \frac{1}{1^3} + \frac{1}{2^3} + \frac{1}{3^3} + \dots + \frac{1}{n^3}, \dots \rightarrow ?? \text{ as } n \rightarrow \infty.$$

$$\sum_{r=1}^{\infty} \frac{1}{r^3} = \zeta(3) \approx 1.20206$$

Convergence

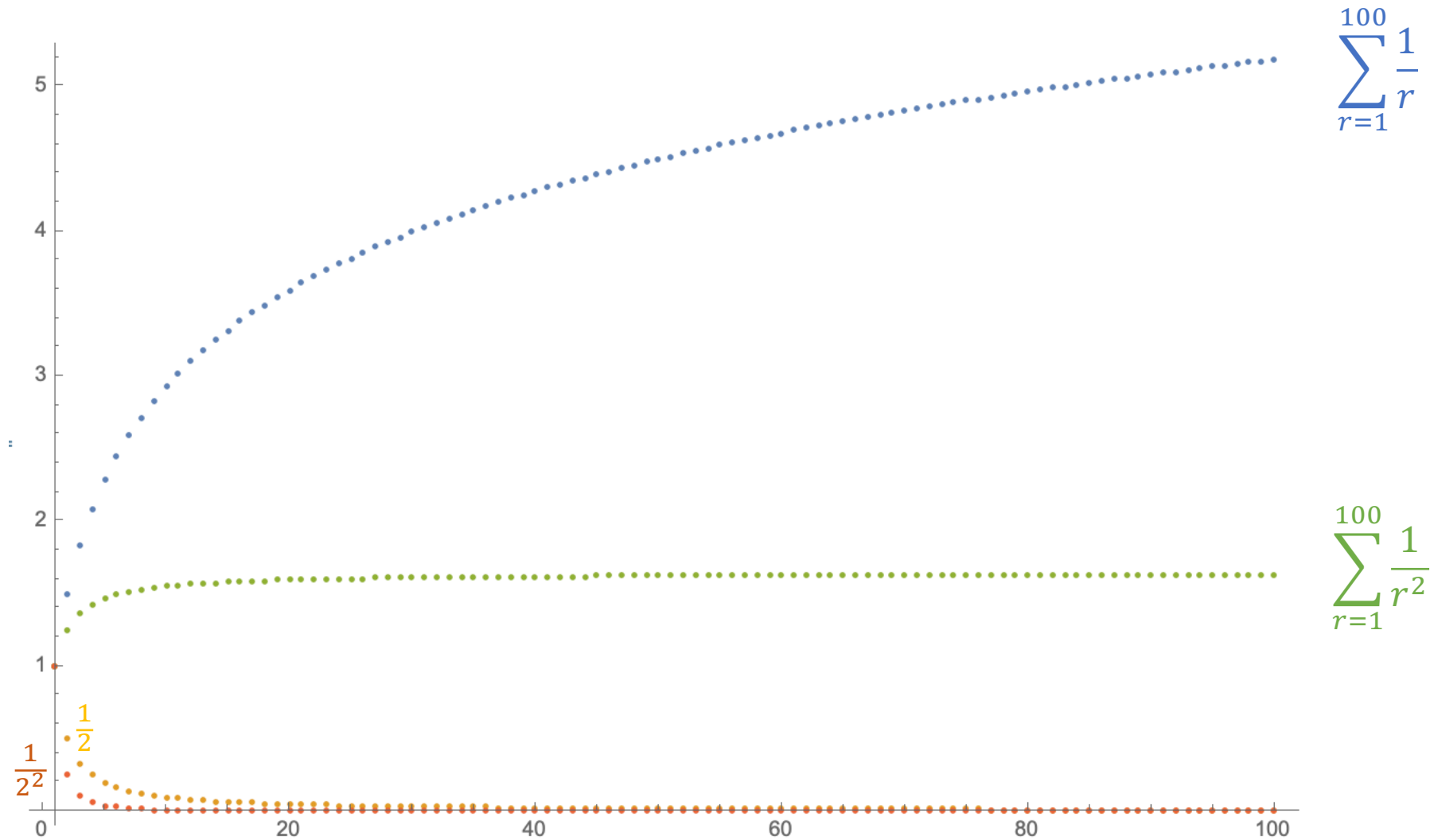
A sequence $\{a_n\}$ **converges to L** if for every $\varepsilon > 0$ there is a natural number N such that, for all natural numbers n ,
If $n > N$, then $|a_n - L| < \varepsilon$.



If a sequence $\{a_n\}$ converges to L we write: $\lim_{n \rightarrow \infty} a_n = L$

A sequence $\{a_n\}$ is said to **converge** if it converges to L for some L , and to **diverge** if it does not converge.

The fine balance of convergence



Convergence for infinite series

We say that an infinite series $\sum_{k=0}^{\infty} a_k$ **converges**, we mean that the sequence of partial sums $S_N = \sum_{k=1}^N a_k$ has a limit as $N \rightarrow \infty$.

If an infinite series does not converge we say it **diverges**.

Note that if a sequence is to converge we must have $\lim_{n \rightarrow \infty} a_k = 0$.

Examples: $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges but $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.

Partial fraction decomposition: simple case

An expression of the form $\frac{ax + b}{(px + q)(rx + s)}$ can be decomposed
into partial fractions of the form $\frac{A}{px + q} + \frac{B}{rx + s}$.

Differencing

Example : Find $\sum_{r=1}^n \frac{1}{r(r+1)}$ and $\sum_{r=1}^{\infty} \frac{1}{r(r+1)}$.

$$\begin{aligned}\sum_{r=1}^n \frac{1}{r(r+1)} &= \frac{1}{r} - \frac{1}{r+1} \\ &= \left(\frac{1}{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1} \right) \\ &= 1 - \frac{1}{n+1} = \frac{n}{n+1}\end{aligned}$$

$$\text{So } \sum_{r=1}^{\infty} \frac{1}{r(r+1)} = \lim_{N \rightarrow \infty} \sum_{r=1}^N \frac{1}{r(r+1)} = \lim_{N \rightarrow \infty} \frac{N}{N+1} = 1.$$

WORKED EXAMPLE 3.5

Find, in terms of n , the sum of $\sum_{r=1}^n \frac{1}{(r+1)(r+2)}$.

Answer

$$\frac{1}{(r+1)(r+2)} = \frac{A}{r+1} + \frac{B}{r+2}$$

Split into partial fractions.

$$1 = A(r+2) + B(r+1)$$

Find the coefficients.

using $r = -1$, $A = 1$ and using $r = -2$, $B = -1$

$$\begin{aligned} \sum_{r=1}^n \left(\frac{1}{r+1} - \frac{1}{r+2} \right) &= \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \left(\frac{1}{4} - \frac{1}{5} \right) + \dots \\ &\quad + \left(\frac{1}{n} - \frac{1}{n+1} \right) + \left(\frac{1}{n+1} - \frac{1}{n+2} \right) \end{aligned}$$

Write down enough terms to see the cancellations.

$$\begin{aligned} &= \left(\frac{1}{2} - \cancel{\frac{1}{3}} \right) + \left(\cancel{\frac{1}{3}} - \cancel{\frac{1}{4}} \right) + \left(\cancel{\frac{1}{4}} - \cancel{\frac{1}{5}} \right) + \dots \\ &\quad + \left(\cancel{\frac{1}{n}} - \cancel{\frac{1}{n+1}} \right) + \left(\cancel{\frac{1}{n+1}} - \frac{1}{n+2} \right) \end{aligned}$$

$$\therefore \sum_{r=1}^n \frac{1}{(r+1)(r+2)} = \frac{1}{2} - \frac{1}{n+2}$$

Cancel terms to get the final answer.

WORKED EXAMPLE 3.6

For the summation $\sum_{r=2}^n \frac{1}{(r-1)(r+1)}$, find an expression in terms of n .

Answer

$$\frac{1}{(r-1)(r+1)} = \frac{A}{r-1} + \frac{B}{r+1}$$

Split into partial fractions.

$$1 = A(r+1) + B(r-1)$$

$$A = \frac{1}{2}, B = -\frac{1}{2}$$

Determine the coefficients.

$$\begin{aligned} \sum_{r=2}^n \frac{1}{(r-1)(r+1)} &= \left(\frac{1}{1} - \frac{1}{3}\right) + \left(\frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \dots \\ &\quad + \left(\frac{1}{n-3} - \frac{1}{n-1}\right) + \left(\frac{1}{n-2} - \frac{1}{n}\right) \\ &\quad + \left(\frac{1}{n-1} - \frac{1}{n+1}\right) \end{aligned}$$

Initially ignore the $\frac{1}{2}$ in front of every term, and note that this sum starts at $r = 2$.

$$= 1 + \frac{1}{2} - \frac{1}{n} - \frac{1}{n+1}$$

Cancel terms.

$$\sum_{r=2}^n \frac{1}{(r-1)(r+1)} = \frac{3}{4} - \frac{1}{2n} - \frac{1}{2(n+1)}$$

Simplify and remember the factor of $\frac{1}{2}$ we previously ignored.

WORKED EXAMPLE 3.7

Given that $S_n = \sum_{r=2}^n \frac{5}{r(r-1)}$, find the sum of the terms as $n \rightarrow \infty$.

Answer

$$\frac{5}{r(r-1)} = \frac{A}{r} + \frac{B}{r-1}$$

$$5 = A(r-1) + Br$$

$$A = -5, B = 5$$

$$\left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n-2} - \frac{1}{n-1}\right) + \left(\frac{1}{n-1} - \frac{1}{n}\right)$$

$$\therefore S_n = \sum_{r=2}^n \frac{5}{r(r-1)} = 5 - \frac{5}{n}$$

$$n \rightarrow \infty, S_n \rightarrow 5$$

Split into partial fractions.

Determine the coefficients.

Initially ignoring the 5 in front of every term, write down enough terms to spot a pattern.

State S_n , remembering to include the 5 we previously ignored.

Write down the limit.



WORKED PAST PAPER QUESTION

Verify that, for all positive values of n ,

$$\frac{1}{(n+2)(2n+3)} - \frac{1}{(n+3)(2n+5)} = \frac{4n+9}{(n+2)(n+3)(2n+3)(2n+5)}.$$

For the series $\sum_{n=1}^N \frac{4n+9}{(n+2)(n+3)(2n+3)(2n+5)}$, find

- the sum to N terms,
- the sum to infinity.

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Answer

$$\begin{aligned} \frac{1}{(n+2)(2n+3)} - \frac{1}{(n+3)(2n+5)} &= \frac{2n^2 + 11n + 15 - (2n^2 + 7n + 6)}{(n+2)(n+3)(2n+3)(2n+5)} \\ &= \frac{4n+9}{(n+2)(n+3)(2n+3)(2n+5)} \end{aligned}$$

$$\begin{aligned} \text{i} \quad \sum_{n=1}^N \frac{4n+9}{(n+2)(n+3)(2n+3)(2n+5)} &= \frac{1}{3 \times 5} - \frac{1}{4 \times 7} + \frac{1}{4 \times 7} - \frac{1}{5 \times 9} + \dots + \frac{1}{(N+2)(2N+3)} - \frac{1}{(N+3)(2N+5)} \\ &= \frac{1}{15} - \frac{1}{(N+3)(2N+5)} \end{aligned}$$

- As $N \rightarrow \infty$, sum becomes $\frac{1}{15}$.

PS 1 Find, in terms of n , an expression for $\sum_{r=2}^n \frac{3}{(r-1)r}$.

P 2 Show that $\sum_{r=1}^n \frac{1}{(r+4)(r+5)}$ can be written as $\frac{n}{5n+25}$.

PS 3 Determine the value of $\sum_{r=1}^{\infty} \frac{1}{(2r+3)(2r+5)}$.

 **PS** 4 Without using a calculator, find, in terms of n , an expression for $\sum_{r=1}^n \frac{1}{(r+3)(r+4)}$.

 **P** 5 Without using a calculator, given that $S_n = \sum_{r=0}^n \frac{1}{(2r+1)(2r+3)}$, show that $S_n = \frac{n+1}{2n+3}$.

 **P PS** 6 Without using a calculator, show that $\sum_{r=3}^n \frac{1}{r(r-2)} = \frac{3}{4} - \frac{1}{2n-2} - \frac{1}{2n}$. Hence, determine, in terms of n ,

 **PS** 7 Without using a calculator, determine the value of $\sum_{r=2}^{\infty} \frac{4}{r^2-1}$.

PS 8 Let $S_n = \sum_{r=4}^n \frac{1}{r^2-5r+6}$.

a Find, in terms of n , an expression for S_n .

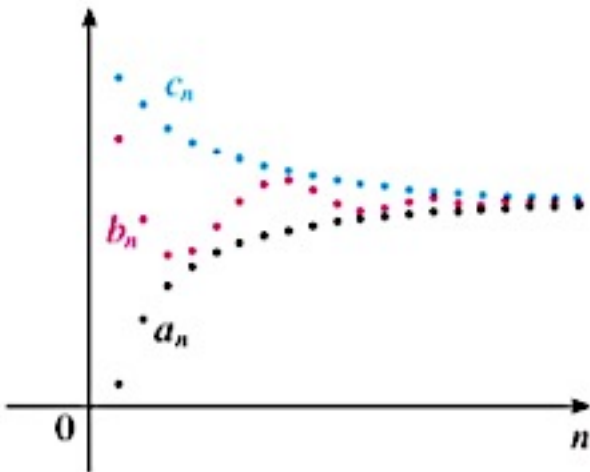
b Given that $S = \lim_{n \rightarrow \infty} S_n$, write down the value of S .

c Find the range of values of n such that $S - S_n < 0.001$.

Stuff about sequences

5 Definition $\lim_{n \rightarrow \infty} a_n = \infty$ means that for every positive number M there is an integer N such that

$$\text{if } n > N \text{ then } a_n > M$$



Squeeze Theorem for Sequences:

If $a_n \leq b_n \leq c_n$ for $n \geq n_0$ and $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$, then $\lim_{n \rightarrow \infty} b_n = L$.

More stuff about sequences

10 Definition A sequence $\{a_n\}$ is called **increasing** if $a_n < a_{n+1}$ for all $n \geq 1$, that is, $a_1 < a_2 < a_3 < \dots$. It is called **decreasing** if $a_n > a_{n+1}$ for all $n \geq 1$. A sequence is **monotonic** if it is either increasing or decreasing.

11 Definition A sequence $\{a_n\}$ is **bounded above** if there is a number M such that

$$a_n \leq M \quad \text{for all } n \geq 1$$

It is **bounded below** if there is a number m such that

$$m \leq a_n \quad \text{for all } n \geq 1$$

If it is bounded above and below, then $\{a_n\}$ is a **bounded sequence**.

12 Monotonic Sequence Theorem Every bounded, monotonic sequence is convergent.

Some challenging sequences

$$\left(\frac{2}{1}\right)^1, \left(\frac{3}{2}\right)^2, \left(\frac{4}{3}\right)^3, \left(\frac{5}{4}\right)^4, \left(\frac{6}{5}\right)^5, \left(\frac{7}{6}\right)^6, \dots \rightarrow e \text{ as } n \rightarrow \infty$$

$$\frac{1}{1} - \ln 1, \quad \frac{1}{1} + \frac{1}{2} - \ln 2, \quad \frac{1}{1} + \frac{1}{2} + \frac{1}{3} - \ln 3, \quad \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} - \ln 4, \dots$$

$$\left(\sum_{r=1}^n \frac{1}{r}\right) - \ln n \rightarrow \gamma \approx 0.577216 \text{ as } n \rightarrow \infty$$

$$1, \quad 1 + \frac{1}{1}, \quad 1 + \frac{1}{1+\frac{1}{1}}, \quad 1 + \frac{1}{1+\frac{1}{1+\frac{1}{1}}}, \quad 1 + \frac{1}{1+\frac{1}{1+\frac{1}{1+\frac{1}{1}}}}, \quad 1 + \frac{1}{1+\frac{1}{1+\frac{1}{1+\frac{1}{1+\frac{1}{1}}}}}, \dots$$

$$\rightarrow \frac{1+\sqrt{5}}{2} \text{ as } n \rightarrow \infty$$

$$\sqrt{\frac{1}{2}}, \quad \sqrt{\frac{1}{2}} \cdot \sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}}, \quad \sqrt{\frac{1}{2}} \cdot \sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}} \cdot \sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}}}, \dots \rightarrow \frac{2}{\pi} \text{ as } n \rightarrow \infty$$

1, 3, 6, 2, 7, 13, 20, 12, 21, 11, 22, 10, ... Is 4 a term in this sequence?

1, 11, 21, 1211, 111221, 312211, 13112221, ... What is the next term in this sequence?

END-OF-CHAPTER REVIEW EXERCISE 3

 Do not use a calculator in this exercise.

-  1 Find the sum of the first n terms of the series $\frac{1}{1 \times 3} + \frac{1}{2 \times 4} + \frac{1}{3 \times 5} + \dots$ and deduce the sum to infinity.

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-  2 Find $2^2 + 4^2 + \dots + (2n)^2$.

Hence find $1^2 - 2^2 + 3^2 - 4^2 + \dots - (2n)^2$, simplifying your answer.

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-  3 Use the method of differences to show that $\sum_{r=1}^N \frac{1}{(2r+1)(2r+3)} = \frac{1}{6} - \frac{1}{2(2N+3)}$.

Deduce that $\sum_{r=N+1}^{2N} \frac{1}{(2r+1)(2r+3)} < \frac{1}{8N}$.

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